

$$Derived_AR \triangleq$$

$$\{ \langle t1, t2 \rangle \in CommittedTxns \times CommittedTxns : \\ tx_commit_time[t1] < tx_commit_time[t2] \\ \}$$

$$Derived_VIS \triangleq$$

$$\{ \langle t1, t2 \rangle \in CommittedTxns \times CommittedTxns : \\ \vee \wedge t2 \neq InitTx \\ \wedge Workload[t2].level \in \{ "RA", "CC", "PC", "PSI", "SI" \} \\ \wedge tx_commit_time[t1] \leq tx_snapshot[t2] \\ \vee \wedge (t2 = InitTx \vee Workload[t2].level = "SER") \\ \wedge \langle t1, t2 \rangle \in Derived_AR \\ \}$$